

Query Retrieval and Evaluation using Vector Space Model

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Abstract

This report describes the final phase of building a simple IR engine. The last phase builds upon the preceding text parsing and indexing work such that this report is continuation or extension of previous report. The query processor development phase enabled document retrieval with Vector Space Model while performing precision and recall-based evaluation. The system employs TF-IDF weighting together with cosine similarity calculation and enables queries through multiple fields that comprise title and description and narrative. The evaluation outcomes show that query breadth levels influence the relationship between precision and recall metrics in the system.

I. Introduction

The final phase of our IR engine includes a query processing module that uses forward and inverted indexes from previous system development stages. The module processes user queries in TREC format by accepting titles and descriptions and narratives. The different query settings emerge through independent or combined usage of these fields. The system analyzes these fields through parsing while conducting tokenization and stop word elimination and stemming in accordance with the indexing preprocessing methods. Query input transforms through weighted TF-IDF vector representation which enables Vector Space Model (VSM) to calculate document-query vector similarity using cosine similarity. The ranking process determines document order according to similarity scores to produce final retrieved results.

The module features an evaluation section which assesses the retrieval effectiveness through its built-in evaluation component. The system evaluates its ranked output through comparison with “main.qrels” which contains human-labeled relevance judgments. The system measures the precision and recall for each query according to three different settings which include title-only queries and title+description queries and title+description+narrative queries to determine the retrieval accuracy of relevant documents. The evaluation method enables researchers to understand query scope trade-offs between precision and recall levels in a retrieval system. The precision rate is typically higher when queries contain only titles yet recall improves when the queries include both descriptions and narratives.

II. System Design

The system operates with three progressive query field scope levels starting from title-only and continuing to title+description and concluding with title+description+narrative. The system handles each version individually for scoring purposes as well as evaluation procedures. The system applies the exact tokenization and stop word deletion and stemming processes to both indexing and query preprocessing. A calculation of weighted TF-IDF

vectors happens for every query version. The forward index automatically builds document vectors that allow the system to compute cosine similarity between query vectors and document vectors. The system retrieves and saves the top-ranked documents using TREC format. The system displays ten top-ranking results for users to verify.

The system evaluates retrieval performance through its ability to analyze different query detail levels in separate modules. Query settings exist independently from each other allowing for systematic evaluation of precision and recall balanced performance. The system protects both precision and memory efficiency through its vector construction method which specifically focuses on accuracy retention.

III. Term Weighting and Normalization

The implemented weighting method for terms utilizes TF-IDF. The TF value equals $(1 + \log_{10}(tf))$ but IDF equals $\log_{10}(N/df)$ where N signifies document number and df represents the term's document occurrences. The model distributes a weight value of 3 to title terms while description terms receive weight 2 and narrative terms receive weight 1. Weight functions apply to primary search terms through this method which still preserves their contextual relationships. Cosine similarity acts as the vector normalization technique.

The system uses multiple weight factors to put direct high-altitude terms first but maintains a balance between important short-specific terms and general semantic information available in long query sections. The produced vectors maintain a comprehensive representation of both the importance ranking of terms and their ability to discriminate between documents. The combination of these factors produces better retrieval outcomes particularly when dealing with extensive or lengthy queries.

IV. Data Structures

The system utilizes Python dictionaries and defaultdicts to create forward and inverted index storage. Each term maps to a dictionary of document IDs and their term frequencies. The sparse retrieval method eliminates every document from the evaluation process unless they contain at least one query term during query processing. The system design enables scalability and operational efficiency with clear interpretability of its procedures.

With this chosen data structure both rapid lookup capabilities and real-time index maintenance become possible in extremely large document systems. The system applies the same calculations to relevant documents alone resulting in reduced execution time and decreased overhead operations. Future system development through query expansion methods and BM25 scoring can be implemented using the straightforward dictionary approach.

V. Evaluation and Results

The retrieval effectiveness metrics precision and recall were computed from relevance judgments in main.qrels for three query settings including title-only and title+description and title+description+narrative. Evaluation of individual query scopes occurred to determine how changing query context dimensions affects retrieval success.:

=== Precision and Recall Comparison ===			
Topic	Setting	Precision	Recall
352	title	0.0040	0.0769
352	title+desc	0.0030	0.0577
352	title+desc+narr	0.0020	0.0385
353	title	0.8750	0.3182
353	title+desc	0.0143	0.4091
353	title+desc+narr	0.0080	0.3636
354	title	0.0896	0.2400
354	title+desc	0.0070	0.2800
354	title+desc+narr	0.0070	0.2800
359	title	0.0011	0.1667
359	title+desc	0.0011	0.1667
359	title+desc+narr	0.0010	0.1667

Fig 1: precision recall for the topics under different settings

🔍 Topic 352 Setting: title	
1.	FT911-3246 – score: 0.2064
2.	FT911-3618 – score: 0.1641
3.	FT911-4701 – score: 0.1449
4.	FT911-1174 – score: 0.1223
5.	FT911-2931 – score: 0.1195
6.	FT911-2267 – score: 0.1179
7.	FT911-3053 – score: 0.1152
8.	FT911-3710 – score: 0.1109
9.	FT911-5027 – score: 0.1056
10.	FT911-4818 – score: 0.1012

Fig 2: sample output ranking for topic352(title).

The research demonstrates marked dissimilarities regarding both topic content and search request parameters. Queries containing only titles achieved more precise results specifically for Topic 353 because its specific search terms matched document semantic content well. The addition of description and narrative fields increased document retrieval in various cases because they retrieved additional pertinent documents beyond those matched by basic query words. Query expansion gives better recall since it retrieves more relevant results yet simultaneously reduces precision because it brings additional non-relevant results into the search results. The query lost focus on its key terms when general vocabulary terms from narrative fields were included in broader topics such as 352 and 359. The number of relevant results decreased strongly while the number of total results improved slightly. The output of query_processor.py files give the retrieved documents for different settings of different topics. They will not be included here due to space constraints.

VI. Conclusion

The query processor functions as the essential final component of the IR engine because it brings together query interpretation with scoring and ranking and evaluation steps. The combination of TF-IDF vectorization with cosine similarity provides organizations with productive methods to create document relevance rankings. The system design enables expandable and understandable query management operations while showcasing conventional IR techniques in organized search systems.

The system achieves its highest precision value by processing focused queries with titles only in Topic 353. The system offers better recall results for exploratory queries because it enables users to include broader context information through description and narrative fields. The observed results indicate that adaptable query weighting techniques together with user-defined scope parameters would optimize actual operational performance. The effectiveness validation phase has established solid groundwork for system development through feedback-based retrieval or semantic matching models.

VII. REFERENCE:

[1] C. D. Manning, P. Raghavan, and H. Schütze, Introduction to Information Retrieval. Cambridge University Press, 2008. [Online].